

VIVEK PATHAK

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

BCA Data Science student at Sri Balaji University, Pune. Hands-on experience in Python, Machine Learning, EDA, Power BI, and SQL through internships, freelance projects, and academic builds. Seeking entry-level roles in Data Science, AI/ML, or Data Analytics.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL

Machine Learning: Scikit-learn, EDA, Feature Engineering, Model Evaluation, SVM, KNN, K-Means, Anomaly Detection

Computer Vision: OpenCV, FaceNet, MTCNN

BI & Visualization: Power BI, DAX, Power Query,

Web & Deployment: Flask, Render, GitHub CI/CD

Databases: PostgreSQL, MySQL

Tools: Jupyter Notebook, GitHub, Excel

EXPERIENCE

Data Analyst Intern | **Bluestock Fintech**

Remote

- Analysed 12 years of financial data for 100 Nifty-listed companies — cleaned inconsistent formats, NULL values, and string anomalies using Python and Pandas
- Built 7 Power BI dashboards with DAX measures covering executive overview, sector comparison, debt monitoring, growth analytics, and dividend tracking
- Developed ML-based Company Health Scoring model (0–100) across 6 financial dimensions — Profitability, Growth, Leverage, Cash Flow, Dividend, Trend
- Applied K-Means clustering, Z-score and Isolation Forest anomaly detection, and revenue trend analysis across all 100 companies
- Key Insight: Identified that free cash flow and 3-year sales CAGR were strongest predictors of overall financial health score

Data Analyst Intern | **CodeFirst Technology**

Remote

- Performed data cleaning, EDA, and validation on structured business datasets using Python and Excel
- Built interactive Power BI dashboards for KPI monitoring and performance trend reporting for stakeholders
- Automated reporting workflows, reducing manual effort and improving turnaround time

AI/ML Freelancer | **ServiceHive**

Remote

- Delivered AutoStream AI Agent — a conversational AI system that classifies social media messages by intent (greeting / product query / high-intent lead) and routes them to lead capture or knowledge-base lookup
- Built with Python, Groq LLaMA 3.3, and FastAPI webhook server; zero hardcoded if-else logic — all routing driven by LLM classification
- Included a JSON knowledge base for product FAQs, pricing, and policies to handle queries without hallucination
- Deliverable: Fully working agent with demo video, tests folder, .env config — independently scoped, built, and delivered

ML Freelancer | **Cognizfy**

Remote

- Completed a structured set of ML tasks across 4 levels covering restaurant data analysis, prediction modelling, and recommendation systems
- Level 1: Data exploration and preprocessing — handled missing values, encoding, outlier treatment on restaurant dataset (9,551 records, 21 features)

- Level 2: Regression model to predict restaurant ratings — applied Linear Regression, Decision Tree, Random Forest; Random Forest achieved lowest RMSE
- Level 3: Built cuisine-type classifier (Logistic Regression, SVM) and customer preference analysis with visualisation
- Level 4: Geospatial analysis of restaurant locations, clustering by city, and a content-based restaurant recommendation engine using feature similarity
- Key Insight: Votes and price range were the top predictors of aggregate rating across all models

Data Analytics Simulation | *Tata & Deloitte Australia via Forage* | *Virtual*

- Tata: EDA, AI-driven delinquency prediction, and business report for collections strategy
- Deloitte: Forensic data analysis, Tableau dashboard, and Excel-based data classification

PROJECTS

Nifty 100 Financial Intelligence Platform — *Python · Pandas · Scikit-learn · Power BI · DAX · PostgreSQL*

- ETL pipeline to ingest and clean 12 years of raw MySQL financial data — standardised year formats, computed 9 derived metrics (D/E, NPM, FCF, Interest Coverage, ROA, Asset Turnover)
- Star-schema PostgreSQL data warehouse powering 7 Power BI dashboards with full drill-through and DAX KPI measures
- ML health scoring model + K-Means clustering to segment 100 companies by financial behaviour; Z-score anomaly detection flagged outlier years
- Insight: HDFC Bank and TCS scored EXCELLENT; high-debt infrastructure companies flagged WEAK — validated against known credit ratings

Real-Time Face Recognition System — *Python · OpenCV · FaceNet · MTCNN · Scikit-learn*

- 3-mode pipeline — Enroll (webcam capture, 30 frames/person), Train (MTCNN detection + FaceNet 512-dim embeddings + SVM/KNN classifier), Run (live recognition with confidence score overlay)
- SVM outperformed KNN on same embeddings; model saved as .pkl for fast reload without retraining
- Handles lighting variation, partial occlusion, multi-face frames; 'Unknown' label applied below configurable confidence threshold (default 0.55)
- Achievement: 95%+ training accuracy; real-time inference at 20+ FPS on standard CPU

AutoStream AI Agent — Social-to-Lead Workflow — *Python · Groq LLaMA 3.3 · FastAPI*

- Conversational AI agent classifies incoming social media messages into 3 intent categories and routes them without hardcoded logic
- JSON knowledge base handles product FAQs and pricing lookups; FastAPI webhook server enables WhatsApp integration
- Delivered for ServiceHive with demo video, test suite, and .env-based LLM provider switching (Groq, OpenAI, Claude, Gemini)

ProTranslator — AI-Powered Translation App — *Python · Flask · Claude AI · Render*

- Full-stack translation app supporting 20 languages with text-to-speech, auto language detection, 2000-char limit, and swap functionality
- Integrated Claude AI REST API as translation engine; deployed on Render with GitHub CI/CD — pro-translator.onrender.com

MBA Eligify — CAT Percentile to B-School Matcher — *Python · Flask · Pandas · Excel*

- Matches CAT percentile and category to 70+ top Indian MBA colleges ranked by ROI and NIRF data; Excel dataset with 30+ attributes per college
- Real-time Pandas filtering with JSON APIs; deployed on Render — mba-eligify.onrender.com

OTT Screen Time & Study Discipline Dashboard — *Power BI · Power Query · DAX*

- Survey of 98 students — avg screen time 2.91 hrs/day, avg study hours 2.17 hrs; DAX measures: Avg Screen Time, Median, Study Hours, Procrastination Index
- Insight: Students with 4+ hrs daily OTT usage scored 40% higher on Procrastination Index; YouTube ranked #1 platform across all demographics

Student Marks Analysis Dashboard — *Power BI*

- Interactive dashboard with class-wise slicers; identified Math and Science as lowest-scoring subjects — used as key recommendation for academic planning

EDUCATION

Bachelor of Computer Applications (BCA) — Data Science Specialisation

Jul 2023 – May 2027

Sri Balaji University, Pune | Coursework: Data Science, ML, Statistics, Python, Data Structures, Data Visualisation

CERTIFICATIONS

- Google — Data Analytics Professional Certificate
- Tata GenAI Powered Data Analytics — Forage
- Deloitte Australia Data Analytics — Forage
- Coursera — Data Analytics & Data Visualizations

ADDITIONAL INFORMATION

Languages: English, Hindi, Marathi

Leadership: Discipline Co-Ordinator (School of Computer Studies), Secretary (Sports Council)

Availability: Open to full-time, internship, and remote roles